Harsh Raj

AI/ML Engineer

Phone: +91 7870259081
Email: harshraj17060@gmail.com
GitHub LinkedIn
Codolio Profile Portfolio

B.Tech. in Mechatronics and Automation Engineering — Indian Institute of Information Technology, Bhagalpur

EDUCATION

Indian Institute of Information Technology, Bhagalpur *2024–2028*
B.Tech. in Mechatronics and Automation Engineering **CGPA: 7.81/10**
Minor in Computer Science and Engineering (CSE), IIIT Bhagalpur
New Holy Ganges Public School, Khagaria, Bihar Class XII: 84%

TECHNICAL SKILLS

Languages: Python, C++

Data Science: Data preprocessing, exploratory data analysis, statistical analysis, feature engineering, data visualization

Machine Learning: Scikit-learn, supervised and unsupervised learning, classification and regression, ensemble methods, reinforcement learning, model evaluation, cross-validation, hyperparameter optimization

Deep Learning: PyTorch, TensorFlow/Keras, neural networks, CNNs, transfer learning

NLP & Computer Vision: Text processing, embeddings, Transformers, OCR, image processing

Generative & Agentic AI: LLMs, LangChain, LangGraph, RAG, prompt engineering, tool calling, multi-agent systems, LLM orchestration, Hugging Face Transformers, SentenceTransformers

Backend, Frontend & Deployment: FastAPI, REST APIs, Pydantic, React, Streamlit, Supabase, SQLite, AWS EC2, Hugging Face Spaces

Testing, Tools & CS Fundamentals: Pytest, unittest, API/integration testing, Git, GitHub, BeautifulSoup, lxml, OOP, DBMS, Operating Systems, Data Structures and Algorithms

PROJECTS

CogniSynth: Multi-Agent Research System *Jun 2026–Present*
Python, LangGraph, Gemini, Tavily, FastAPI, Streamlit GitHub | Live Demo

- Built a LangGraph-based research workflow orchestrating **6 active agents** to decompose topics, research web sources, extract evidence, verify claims, synthesize findings, and generate reports.
- Implemented evidence-linked claim verification and contradiction detection; processed **41 claims** and identified **13 contradictions** across 4 production runs.
- Integrated FastAPI/SSE streaming, telemetry, and local TF-IDF semantic memory with **under 5 ms top-4 retrieval**; telemetry showed an estimated mean LLM cost of **$0.00249 per completed run** (n=4).

Brand AI-Readiness Auditing Engine *Aug 2026–Present*
Python, Web Crawling, HTML Parsing, JSON, FastAPI, React GitHub | Live Demo

- Built a FastAPI/React website-auditing engine spanning **11 AI-readiness categories** and **35 distinct checks** (24 test-covered), with 0–100 citability scoring and automated Schema.org/robots.txt remediation snippets; achieved **94.4% audit completion** across 18 real-world domains.
- Implemented bounded, robots-aware crawling and multi-tier SSRF safeguards; DOM/schema signal extraction averaged **0.272 ms**, while unsafe-address rejection averaged **1.37 ms** across 14 test patterns.
- Revalidated **23/23 API tests**, **45/45 pre-deployment checks**, **13/13 self-test fixtures** (123 check points), and all 6 multi-skill integrations; the frontend production build compiled 1,978 modules.

MediReach: AI-Enabled Healthcare Platform *2026*
Team Project | LLM Services, OCR, Document Processing, FastAPI GitHub

- Built an LLM-driven patient-intake workflow to collect and structure patient information, with a document pipeline to extract data from uploaded medical records.
- Developed AI-assisted report generation and protocol-oriented review; an LLM reviewer flags contradictions and unsupported claims in generated reports.
- Integrated AI services with FastAPI and contributed to SQLite persistence, API-key authentication, configurable multilingual support, medicine-label OCR, and voice-assistance workflows; team secured second place at the SIH '26 internal hackathon.

ACHIEVEMENTS AND CERTIFICATION

View Certificates

- **2nd Place**, Synapse Arena (Reinforcement Learning-based Racing Agent Challenge), PROTOFOS Tech Fest '26, IIIT Bhagalpur.
- **3rd Place**, AI-Verse '26 Machine Learning Challenge, IIIT Bhagalpur (2026).
- **2nd Place**, SIH '26 Internal Hackathon (2026).
- Solved **200+ DSA problems**; profile: Codolio.
- **Introduction to Large Language Models**, IBM SkillsBuild, Jan 2026.

LEADERSHIP

Project Manager, Robotics and Electronics Club, IIIT Bhagalpur *2024–2025*

- Coordinated about **5 teams** across **4 robotics projects**; organized **10+ technical workshops** for an average of **25 students**, mentored **20+ students**, and contributed to management of Enyugma, IIIT Bhagalpur's techno-cultural festival.